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Vaginal microbiota alterations under supraphysiological estradiol state during in vitro fertilization-embryo transfer (IVF-ET) and the association with reproductive outcomes

Xijing Chen^{1*†}, Feng Chen^{1†}, Shufang Wu¹, Pingping Lv¹ and Ping Li^{2*}

Abstract

Background The process of in vitro fertilization-embryo transfer (IVF-ET) induces a maternal supraphysiological estradiol environment during embryo implantation and early development. Estrogen is crucial in modulating the colonization of microbiota within the vaginal epithelium. However, the impact of supraphysiological estradiol levels on the vaginal microbiome and the relationship with pregnancy outcomes remains unclear.

Results The study aimed to characterize the vaginal microbiota under supraphysiological hormonal conditions. A total of 67 patients undergoing fresh embryo transfer were divided into three groups based on their peak estradiol levels: high-estradiol (HE) group ($E_2 > 11,000$ pmol/L), median-estradiol (ME) group ($E_2 5,000\text{--}11,000$ pmol/L), and low-estradiol (LE) group ($E_2 < 5,000$ pmol/L). Twenty-five patients undergoing frozen-thawed embryo transfer were categorized into natural cycle (NC) group and hormone replacement cycle (HRT) group according to endometrial preparation protocols. Using 16S rRNA sequencing, we found that the vaginal microbiome exhibited variations with changes in peak estradiol levels. The elevated estradiol levels during ovarian stimulation or exogenous estrogen supplementation, significantly reduced alpha diversity, altered beta diversity within the vaginal microbiome, and shifted the vaginal community state types (CSTs) in Chinese infertile women toward *Lactobacillus*-dominant profiles, resembling those observed in most Asian women previously. However, the reproductive outcomes were not improved by these variations. The *Streptococcus_anginosus* and *Akkermansia* abundance correlated with estradiol levels positively, whereas *Escherichia-Shigella* showed a negative correlation. The abundance of *Streptococcus*, *Atopobium*, and *Bifidobacterium* on the day of embryo transfer may serve as predictors for adverse pregnancy outcomes, as determined by calculating the area under the curve (AUC) values.

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Conclusions Supraphysiological estradiol levels induced by IVF-ET significantly alter vaginal microbiota and shift the CSTs in Chinese infertile women toward patterns of most Asian women. The *Lactobacillus* dominance under supraphysiological estradiol conditions does not help improve assisted reproductive outcomes. The abundance of *Streptococcus*, *Atopobium*, and *Bifidobacterium* on the day of embryo transfer may serve as predictors for adverse pregnancy outcomes. Among them, *Streptococcus* correlates positively with peak estradiol levels and may act as a microbial mediator impairing reproductive success under hyperestrogenic conditions. However, further larger-scale researches are needed to identify and elucidate the potential mechanisms.

Keywords Vaginal microbiome, In vitro fertilization-embryo transfer (IVF-ET), Supraphysiological estradiol (E₂) level, Pregnancy outcomes

Background

Microorganisms adhere to various surfaces and cavities within the human body, including the gut, skin, oral cavity, and vagina [1]. The microbial inhabitants of the vagina exist in a mutualistic relationship with the host, playing a critical role in maintaining vaginal health [1]. A healthy vaginal microbiome is typically characterized by a dominance of *Lactobacillus* coupled with a low diversity of bacterial species [2]. *Lactobacillus* can produce antimicrobial peptides such as bacteriocins, bacteriocin-like substances, and biosurfactants, and can also promote autophagy in intracellular bacteria, viruses, and protozoa [3]. In conjunction with their antimicrobial and anti-inflammatory products, *Lactobacillus* interacts with components of the epithelial mucosal barrier to provide the first line of defense against pathogen invasion [3]. A *Lactobacillus*-dominated microbiome profile has been found to be positively correlated with reproductive success, whereas bacterial vaginosis is associated with poorer results [2].

The colonization of the female reproductive tract by *Lactobacillus* is associated with estrogen levels [1]. Elevated estrogen levels can stimulate the proliferation of vaginal epithelial cells and promote the maturation and deposition of glycogen within these cells [3]. Glycogen from exfoliated epithelial cells is catabolized into smaller polymers in the vagina and subsequently metabolized into lactic acid by *Lactobacillus*. This lactic acid stimulates the dissolution of epithelial cells and promotes the accumulation of glycogen. The resulting acidic vaginal environment favors the growth of *Lactobacillus* and inhibits the proliferation of other organisms [1, 3]. The fluctuating estrogen levels throughout the menstrual cycle contribute to the dynamic variation in vaginal microbiota. The interplay between estrogen levels, epithelial glycogen, and *Lactobacillus* helps maintain an acidic vaginal microenvironment in reproductive-age women [1].

The in vitro fertilization-embryo transfer (IVF-ET) process can induce a maternal supraphysiological estradiol environment during embryo implantation and early development [4]. Supraphysiological estradiol levels may arise from either the development of multiple

follicles during controlled ovarian stimulation (COS), leading to estradiol concentrations 10 to 20 times higher than physiological levels, or from exogenous estradiol supplementation for endometrial preparation in frozen-thawed embryo transfer (FET) cycles, creating a non-physiological maternal endocrine environment [5]. Estrogen can influence the colonization of microbiota in vaginal epithelium and the female susceptibility to pathogenic microbial infections [3, 6]. Recently, vaginal microbial diversity is found to be one of the important factors for achieving good prognosis in IVF-ET [7–9], while excessively high estradiol levels are also thought to have adverse effects on pregnancy outcomes [4, 10–13]. However, whether the supraphysiological estradiol interferes with the vaginal microbiome and negatively affects IVF outcomes remains inconclusive. Our study aims to examine the compositional characteristics of the vaginal microbiota in different estrogen milieus induced by IVF treatment and further explore the relationships among supraphysiological estradiol levels, the vaginal microbiome and the pregnancy outcomes.

Methods

Subject recruitment

A prospective cohort study was conducted at the Assisted Reproductive Technology (ART) Center of Women's Hospital, School of Medicine, Zhejiang University, from April 2023 to March 2024, with the approval of the Hospital Ethics Committee (ID: IRB-20250024-R) (Fig. S1). The study enrolled 92 women, aged 22–40 years, diagnosed with infertility and indicated for IVF or intracytoplasmic sperm injection (ICSI). All participants were transferred two high-grade cleavage stage embryos. Exclusion criteria included a history of recurrent spontaneous abortion (including ≥ 2 times of biochemical pregnancies), recurrent implantation failure, untreated hydrosalpinx, untreated uterine abnormalities, stage III/IV endometriosis or adenomyosis, and acute reproductive tract infection. Patients who had used antibiotics, vaginal probiotics, or immunomodulatory medication during the embryo transfer cycle were also excluded.

In vitro fertilization treatment and follow-up

Standardized operating procedures were followed during the IVF-ET procedures. The gonadotropin-releasing hormone (GnRH) antagonist regimen was used for ovarian stimulation in patients undergoing fresh embryo transfer. Gonadotropin, at a dose of 100–300 IU per day, was administered on day 2–3 of the menstrual cycle. A GnRH antagonist dose of 0.25 mg daily was initiated from day 5–6 of ovarian stimulation or when at least one dominant follicle reached 12 mm in diameter. Ovarian response was monitored, and the gonadotropin dose was adjusted until at least two dominant follicles reached 18 mm in diameter, or at least three reached 17 mm. Human chorionic gonadotropin (hCG) +/- GnRH agonist was administered for trigger, and oocyte retrieval was performed 35.5 to 37 h later. IVF or ICSI was selected based on the indications. On day 3 post-oocyte retrieval, two high-grade embryos were transferred in suitable patients. Luteal phase support was initiated the following day of oocyte retrieval using vaginal progesterone gel and oral dydrogesterone. Ten to 14 days post-embryo transfer, blood hCG levels were measured. If the hCG test was positive, the patient should be followed-up. Luteal phase support was continued until approximately 10 weeks of gestation in cases where viable intrauterine pregnancies were diagnosed.

For patients undergoing frozen-thawed embryo transfer, natural ovulatory cycle or hormone replacement cycle regimen was used for endometrial preparation. In natural ovulatory cycle, follicle monitoring was initiated from day 10–12 of the menstrual cycle until ovulation. In hormone replacement cycle, estradiol valerate was taken orally from the 2nd to 3rd day of menstruation for at least 12 days, and the endometrium was transformed with progesterone. Two frozen cleavage stage embryos were thawed and transferred on the 3rd day post-ovulation or progesterone administration. Luteal phase support was administered and continued until 10 weeks of gestation if pregnancy was achieved. The pregnancy outcomes of all patients were followed up.

Participants undergoing fresh embryo transfer were divided into three groups based on the estradiol level on the trigger day, designated as high-estradiol (HE) group with the estradiol levels > 11,000 pmol/L, median-estradiol (ME) group with the estradiol levels ranged from 5,000 to 11,000 pmol/L, and low-estradiol (LE) group with the estradiol levels < 5,000 pmol/L. Those undergoing frozen-thawed embryo transfer were designated as the natural cycle (NC) group or the hormone replacement cycle (HRT) group.

Sample collection

Vaginal samples were collected pre-embryo transfer using a sterile speculum and cotton swab, which was

gently rotated across the upper 1/3 of the vaginal wall and the posterior fornix. The samples were stored at – 80 °C for future use [14].

DNA extraction and 16S rRNA sequencing

Total DNA was extracted from the collected samples utilizing a commercial genome DNA extraction kit (DP302, Tiangen Company, Beijing, China), according to the manufacturer's instructions. Total DNA served as the template for PCR amplification, employing the universal primer 341F/805R (341F: 5'-CCTACGGGNGGCWGCAG-3'; 805R: 5'-GACTACHVGGGTATCTAATCC-3') under the following conditions: pre-denaturation at 98 °C for 30 s, denaturation at 98 °C for 10 s, annealing at 54 °C for 30 s, extension at 72 °C for 45 s, and 32 cycles, with a final extension at 72 °C for 10 min. The PCR product was purified, quantified, and then assessed using an Agilent 2100 Bioanalyzer (Agilent, Santa Clara, CA, USA) and Illumina library quantitative kits (Kapa Biosciences, Woburn, MA, USA). Qualified PCR products were pooled together and sequenced on the Illumina NovaSeq 6000 platform (Paired-End 250) [15], with sequencing services provided by LC-Bio Technology Co., Ltd., Hangzhou, China.

Data processing and 16S rRNA sequencing analysis

Paired-end reads were merged after removing the sequencing primer from demultiplexed raw sequences. Low-quality reads (quality scores < 20), short reads (< 100 bp), and reads containing more than 5% "N" records were trimmed to obtain high-quality clean data. The Divisive Amplicon Denoising Algorithm (DADA2) was applied for denoising and generating amplicon sequence variants (ASVs). Taxonomic annotation of representative sequences was performed using the QIIME2 plugin feature-classifier (version 2019.7, <https://qiime2.org/>), with reference to the SILVA and NT-16S databases [15, 16].

Bioinformatics and statistical analysis

Alpha and beta diversities were calculated using QIIME2 [16], with relative abundance applied in bacteria taxonomy. Alpha diversity was measured using Shannon's diversity index, Simpson's diversity index, observed species and Chao1 richness estimator. Kruskal-Wallis test was used to compare the difference between groups. Beta diversity was evaluated using jaccard distance matrices and represented as principal coordinates analysis (PCoA), with the significance among groups determined by Adonis.

The Kruskal-Wallis test was used to identify differentially abundant genera, with the significance set at $p < 0.05$. The linear discriminant analysis (LDA) effect size

(LEfSe) was applied to identify the microbial biomarkers among groups ($LDA \geq 2.0$, $p < 0.05$) [17].

Spearman's correlation analysis was conducted to examine the correlation between the estradiol levels and the relative abundance of vaginal microbiota. Spearman's rho (r) and the p value were calculated.

Receiver operating characteristic curves (ROC) were generated to evaluate the predictive value of microbial abundance in pregnancy outcomes. Biomarkers with the area under the ROC curve (AUC) greater than 0.7 were considered meaningful.

Fundamental statistical analyses were performed using SPSS 23.0 software. The normality of all continuous variables was assessed through descriptive statistics for skewness and kurtosis, visual inspection of histograms, and the Kolmogorov-Smirnov test. Non-normally distributed continuous variables were presented as the median (25th–75th percentile, P25–P75), while categorical variables were expressed as percentages. Group comparisons were conducted using the Kruskal-Wallis test or Chi-square test, as appropriate. A p value of < 0.05 was considered statistically significant.

Results

General clinical characteristics of the participants

A total of 92 patients received embryo transfer were enrolled in the study. They were divided into five groups according to the estradiol levels on the trigger day before fresh embryo transfer or the endometrium preparing regimens before frozen embryo transfer. There were 15 patients in the NC group, 10 in the HRT group, 16 in the LE group, 26 in the ME group and 25 in the HE group respectively.

There were no significant differences between the five groups in terms of age, body mass index (BMI), history of ectopic pregnancy or inevitable abortion, fertilization strategy (IVF/ICSI), or preimplantation endometrial thickness (Table 1). However, the percentage of patients with diminished ovarian reserve (DOR) was significantly higher in the LE group compared to the HE group ($p < 0.001$). No significant differences were observed among the five groups regarding other etiologies of infertility. The duration of infertility was significantly longer in the NC and HRT groups compared to the other groups ($p = 0.004$). The median anti-müllerian hormone (AMH) value in the LE group was the lowest, followed by the ME group, which was significantly lower than in the other three groups ($p < 0.001$). The peak estradiol value was highest in the HE group, followed by the ME, LE, HRT, and NC groups ($p < 0.001$), with no significant difference between NC and HRT group. Data is presented in Table 1.

Diversity and composition of vaginal microbial community

The comparison of alpha diversity among the five groups is presented in Fig. 1A. The Shannon's diversity index was significantly higher in the LE group compared to the ME, HE, and HRT groups ($p < 0.05$), with no difference observed among the other groups. These results indicate that groups with higher peak estradiol levels or the addition of exogenous estrogen exhibited lower vaginal microbial diversity compared to the LE group.

PCoA was applied to illustrate the distribution of vaginal microbial community across the different groups. By calculating the jaccard distance matrices, we identified that the LE group was distinct from the HRT group in terms of beta diversity (Adonis $p < 0.05$), while there was no significant difference among the other groups (Fig. 1B). The findings suggested that the microbial community structure was similar among most groups, with the exception of the LE and HRT groups.

To characterize the vaginal microbial community taxonomy across different groups, we further calculated the relative abundance of vaginal microbiota at the phylum and the genus levels. At the phylum level, the top four dominant groups were Bacillota, Actinomycetota, Pseudomonadota and Bacteroidota. In the fresh embryo transfer groups, as the peak of estradiol increased, the proportion of Bacillota increased. In the frozen embryo transfer groups, the HRT group receiving exogenous estrogen supplementation exhibited a higher Bacillota proportion compared to the NC group with physiological estradiol concentrations. In the LE and NC groups characterized by lower estradiol peaks, Pseudomonadota demonstrated greater relative abundance (Fig. 1C).

At the genus level, the top four dominant groups were *Lactobacillus*, *Gardnerella*, *Streptococcus* and *Atopobium*. In patients undergoing fresh embryo transfer, the dominance of *Lactobacillus* became more pronounced as the estradiol peak levels on the trigger day increased. The HE group, with the highest estradiol peak, had the highest average proportion of *Lactobacillus* (68.93%), while the LE group, with the lowest estradiol peak, had the lowest proportion (53.04%), which was similar to that in the NC group (53.67%) receiving freeze-thaw embryo transfer. In patients receiving frozen embryo transfer, the average *Lactobacillus* proportion in the HRT group was higher than that in the NC group, and was similar to the HE group receiving fresh embryo transfer. However, no significant differences in *Lactobacillus* abundance were observed among the groups. The highest abundance of the potentially pathogenic bacterium *Streptococcus* and *Atopobium* was seen in the HE group and the HRT group respectively, although the differences were not statistically significant (Fig. 1D).

Table 1 Clinical characteristics of participants

	NC (n = 15)	HRT (n = 10)	LE (n = 16)	ME (n=26)	HE (n=25)	p value
Age (years old)	34 (30-39)	32.5 (28.5-35)	31.5 (29-34.75)	31.5(28.75-34.25)	32(28-34.5)	0.377
BMI (kg/m ²)	20.24 (18.1-24.84)	23.655 (20.535-25.7575)	22.12 (19.345-24.38)	21.6(20.3475-25.1125)	21.09(19.84-23.31)	0.470
Etiology of infertility _n (%)						<0.001
Tubal infertility	6 (40)	6(60)	5 (31.3)	12(46.2)	16(64)	
Male infertility	4 (26.7)	2(20)	3 (18.8)	10(38.5)	5(20)	
Diminished ovarian reserve (DOR)	2 (13.3) ^{ab}	0(0) ^{ab}	7 (43.8) ^b	2(7.7) ^{ab}	0(0) ^a	
Unexplained infertility	3 (20)	0(0)	1 (6.3)	2(7.7)	4(16)	
Anovulation	0 (0)	2(20)	0 (0)	0(0)	0(0)	
History of ectopic pregnancy _n (%)	2 (13.3)	1(10)	3 (18.8)	6(23.1)	8(32)	0.543
History of inevitable abortion _n (%)	5 (33.3)	0(0)	1 (6.3)	6(23.1)	2(8.0)	0.065
Fertilization strategy _n (%)						0.433
IVF	9 (60)	8 (80)	13 (81.25)	18(69.2)	21(84)	
ICSI	6 (40)	2 (20)	3 (18.75)	8(30.8)	4(16)	
AMH (ng/ml)	3.24 (2.3425-4.3050) ^a	4.38 (2.295-9.5925) ^a	1.3 (1.03-1.92) ^b	2.01(1.44-2.9075) ^c	3.38(2.59-4.365) ^a	<0.001
Duration of infertility (years)	5 (2-6) ^a	3.5 (2-5.25) ^{ac}	1 (1-3) ^b	1.75(1-3) ^b	2(1-4) ^{bc}	0.004
Endometrial thickness (cm)	1.03 (1.0-1.12)	1.015 (0.875-1.14)	1.115 (1.0125-1.27)	1.09(0.9375-1.275)	1.15(1.005-1.235)	0.127
The peak of estradiol (pmol)	1076 (640-1355) ^a	1312 (795.5-2158.25) ^{ab}	4299 (3526.5-4789) ^b	7744(6155.25-9424) ^c	12,953(12303-14302) ^d	<0.001
Clinical outcome _n (%)						0.556
Ongoing pregnancy	14 (93.3)	6 (60)	11 (68.75)	21(80.77)	18(72)	
Biochemical pregnancy	0 (0)	0 (0)	1 (6.25)	1(3.85)	3(12)	
Un-pregnancy	1 (6.7)	3 (30)	4 (25)	3(11.53)	3(12)	
Early abortion	0 (0)	1 (10)	0 (0)	1(3.85)	1(4)	
Live birth _n	14	6	11	20	18	
Singleton birth _n (%)	11 (78.6)	6 (100)	10 (90.9)	12(60)	11(61.1)	0.129
Twin birth _n (%)	3 (21.4)	0 (0)	1 (9.1)	8(40)	7(38.9)	

Variables are presented as median (P25-P75) (Kruskal-Wallis test) or n (%) (Chi-square test). Statistical significance is considered at $p < 0.05$. Superscript letters a, b, c, d are used to represent the significance of differences between groups. Different letters indicate a significant difference between the groups, while the same letter indicates no significant difference.

The distribution of microbial communities

We employed the item community state types (CSTs) proposed by Ravel J et al. in 2011 to stratify the vaginal microbiota in women with reproductive age and compared the distribution of microbial community types across the five groups. CST I, II, III, V are dominated by *Lactobacillus crispatus*, *Lactobacillus gasseri*, *Lactobacillus iners* and *Lactobacillus jensenii* respectively, while CST IV is characterized by higher proportions of strictly anaerobic organisms [18]. CSTs I, III, IV, and V were observed in the participants of the study. In the NC group, CST IV accounted for the highest proportion (40%), while CSTs I, III, and V each accounted for 20%. In the HRT group, CST III had the highest proportion (60%), followed by CST IV (20%), with CSTs I and V each accounting for 10%. The distribution of CSTs in the LE group was similar to that of the NC group, with CST IV

accounting for the highest proportion (43.75%), followed by CST I (31.25%), and CST III and CST V each accounting for 12.5%. The ME group had the highest proportion of CST III (42.3%), followed by CST IV (30.8%), CST I (23.1%), and CST V (3.8%). Similarly, the HE group also had the highest proportion of CST III (44%), followed by CST I (32%), CST IV (20%), and CST V (4%). However, the distribution of microbial communities did not significantly differ among the groups ($p = 0.290$) (Fig. 2).

Differentially abundant species across the groups

We conducted LEfSe analysis to identify the microbial biomarkers and analyzed the differentially abundant species among the groups, data was shown in Fig. 3; Table 2. At the genus level, *Escherichia-Shigella* was identified as the biomarker for the HRT group, with its abundance significantly higher in the HRT group compared to the ME

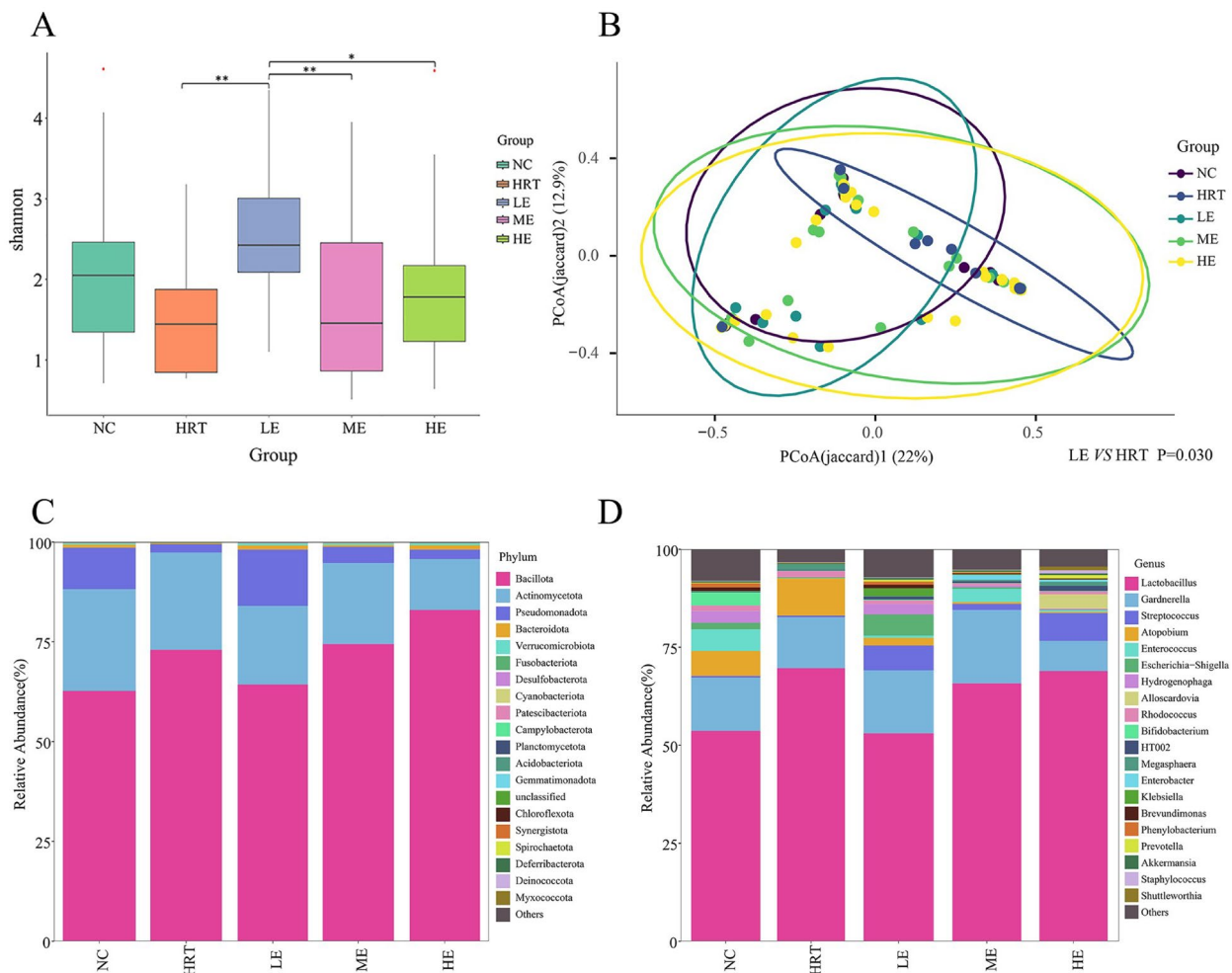


Fig. 1 The diversity and taxonomy of vaginal microbiota among groups with different peak estradiol levels. (NC: natural cycle group; HRT: hormone replacement cycle group; LE: low-estradiol group; ME: median-estradiol group; HE: high-estradiol group). **(A)** Comparison of Shannon diversity indexes of vaginal microbiota. The p value was determined by the Kruskal-Wallis test. * $p < 0.05$, ** $p < 0.01$. **(B)** Principal coordinates analysis (PCoA) of vaginal microbial community structure based on jaccard distance (Adonis $p = 0.030$). **(C)** Distribution of vaginal bacterial composition at the phylum level. **(D)** Distribution of vaginal bacterial composition at the genus level. The top 20 bacterial species are displayed. The horizontal axis represents different groups. The vertical axis represents the relative abundance of vaginal microbiota

and HE groups ($p = 0.027$), but no significant difference was observed when compared to the LE or NC groups. *Prevotella* was the biomarker for the LE group, with significantly higher abundance in the LE group compared to the ME, NC, and HRT groups ($p = 0.003$). *Akkermansia* was the biomarker for the HE group, and its abundance was significantly lower in the HRT group than in the other groups ($p = 0.001$).

At the species level, *Streptococcus anginosus* was the biomarker for the ME group, with its abundance highest in the ME group, similar to the levels observed in the HE and LE groups, but significantly higher than in the NC and HRT groups ($p = 0.003$). *Lactobacillus iners* was identified as the biomarker for the HRT group. Its abundance was highest in the HRT group, followed by the ME and HE groups, with no significant difference among the

three groups. However, all three groups had significantly higher abundance than the NC and LE groups ($p = 0.031$).

Correlation between peak estradiol levels and the vaginal microbiota

Given that the study grouped the subjects based on estradiol peaks, we further explored the correlation between peak estradiol levels and the vaginal microbiota (Table 3). At the genus level, the abundance of *Escherichia-Shigella* was negatively correlated with peak estradiol ($p = 0.005$), while the abundance of *Akkermansia* was positively correlated ($p = 0.001$). At the species level, the abundance of *Streptococcus anginosus* was positively correlated with peak estradiol levels ($p = 0.026$).

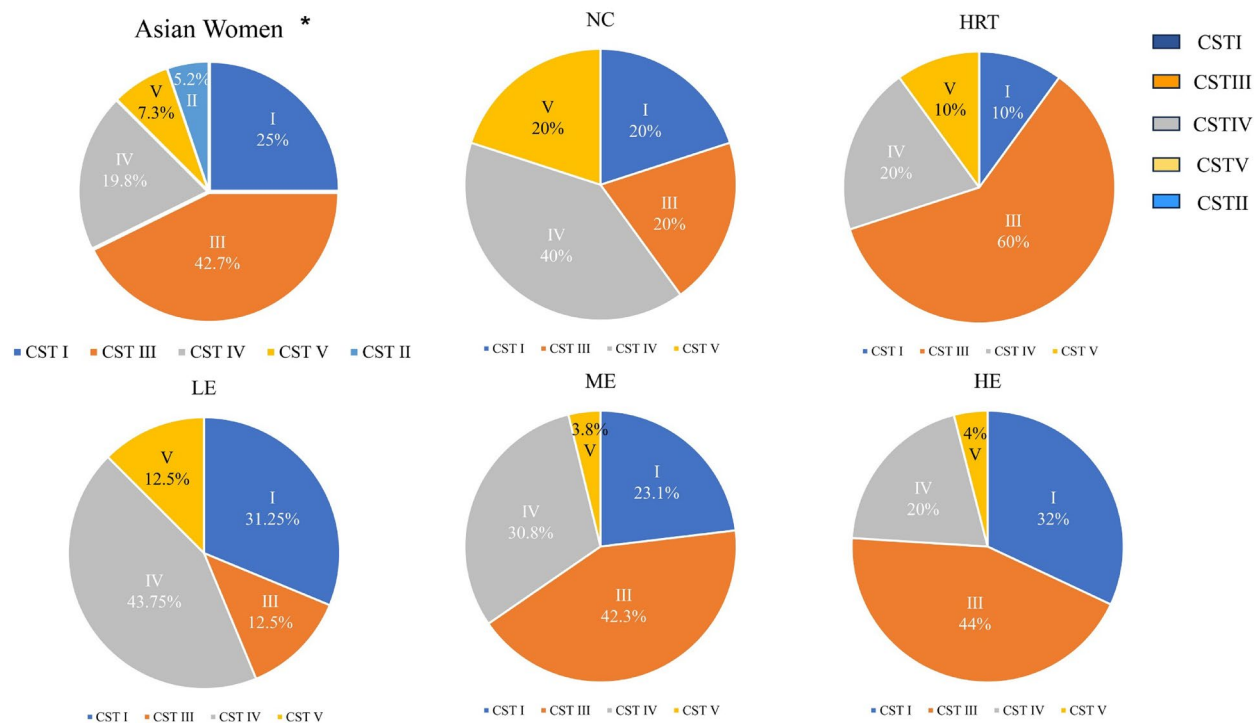


Fig. 2 The distribution of microbial community state types (CSTs) between Asian women and five study groups. (NC: natural cycle group; HRT: hormone replacement cycle group; LE: low-estradiol group; ME: median-estradiol group; HE: high-estradiol group) No significant difference was detected among the five groups ($p=0.290$) *The distribution map of CSTs in Asian women was replotted based on data from Ravel J et al. [18]

Pregnancy outcomes

The ongoing pregnancy rate was highest in the NC group (93.3%), followed by the ME group (80.77%), HE group (72%), LE group (68.75%), and HRT group (60%). However, there was no significant difference in pregnancy outcomes among the five groups ($p=0.556$), as shown in Table 1.

Biochemical pregnancies occurred only in the three groups that underwent fresh embryo transfer (HE, ME, and LE groups), with proportions of 12%, 3.85%, and 6.25%, respectively. All abortions took place during the first trimester. Early abortions were reported in the HRT, HE, and ME groups, with proportions of 10%, 4%, and 3.85%, respectively. Except for one case where the pregnancy was voluntarily terminated due to maternal factors, all other ongoing pregnancies resulted in live births, including 50 singleton births (72.5%) and 19 twin births (27.5%). There was no significant difference in the proportion of twin births among the five groups ($p=0.129$).

Analysis of the vaginal microbiota in relation to pregnancy outcomes

We further categorized all samples into four groups based on pregnancy outcomes: the ongoing pregnancy (OP) group, biochemical pregnancy (BP) group, un-pregnancy (UP) group and abortion (AB) group. There were no significant differences among the four groups in terms

of age, BMI, etiology of infertility, duration of infertility, history of ectopic pregnancy or inevitable abortion, fertilization strategy (IVF/ICSI), peak estradiol values, or preimplantation endometrial thickness. Species differences across different pregnancy outcomes were analyzed, and LEfSe analysis was used to identify biomarkers. At the genus level, *Streptococcus* was identified as the biomarker for the UP group, with its abundance significantly higher in the UP group compared to the OP group ($p=0.003$). *Atopobium* was the biomarker for the AB group, with significantly higher abundance in the AB group than in the OP group ($p=0.040$). *Bifidobacterium* was identified as the biomarker for the BP group, with its abundance significantly higher in the BP group compared to the OP and UP groups ($p=0.019$). The data are presented in Fig. 4; Table 4.

Evaluation of the predictive accuracy of the vaginal microbiota for pregnancy outcomes

To evaluate the potential predictive value of the vaginal microbiota on the pregnancy outcomes, we plotted ROC curves to distinguish among different pregnancy outcomes (Fig. 5). The AUC for *Streptococcus* in predicting non-ongoing pregnancy was 0.758, with a 95% confidence interval (CI) ranging from 0.658 to 0.858 ($p<0.001$). Similarly, the AUC for *Streptococcus* in predicting non-pregnancy was 0.764, with a 95% CI ranging from 0.662

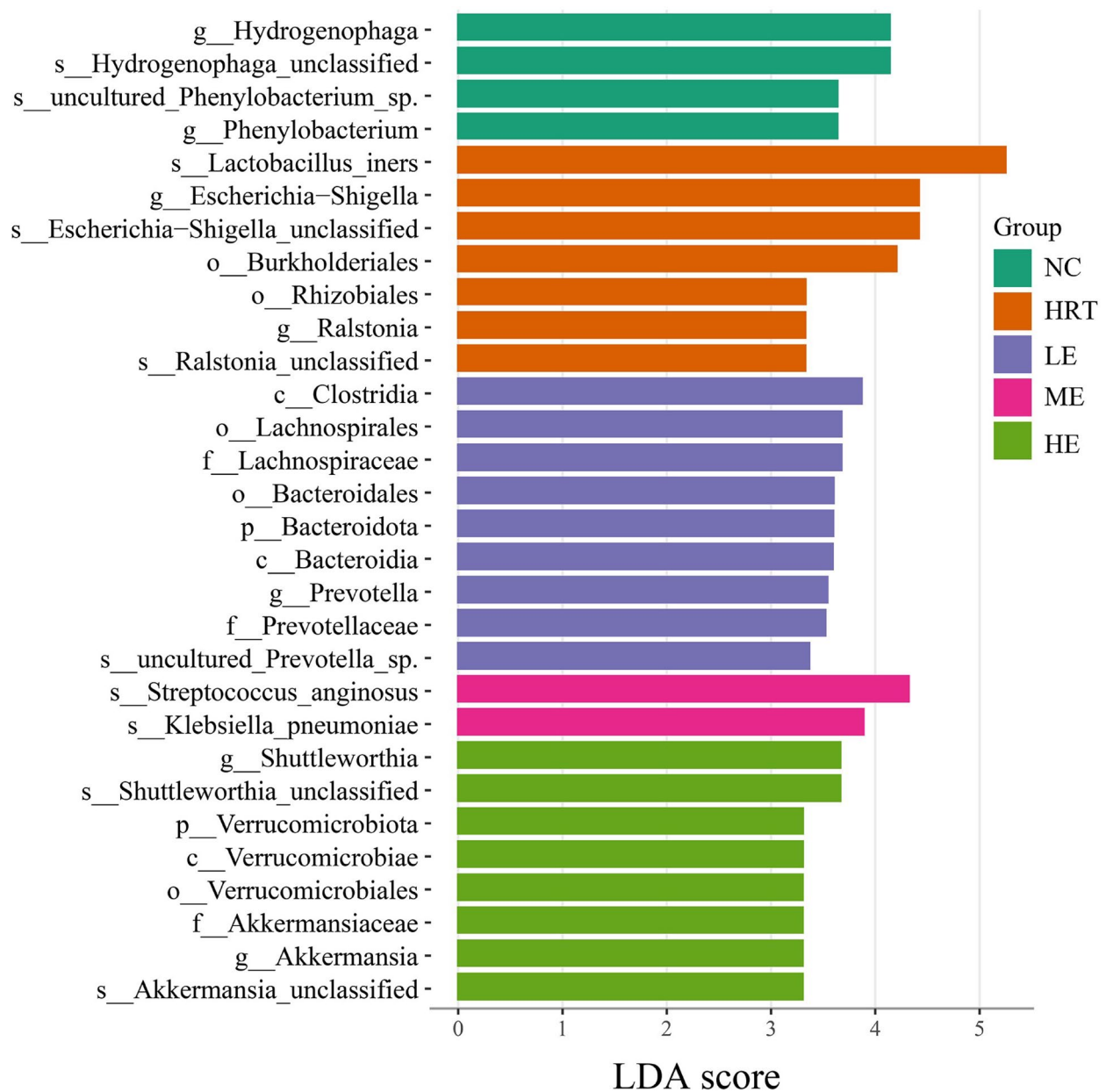


Fig. 3 Differentially abundant species across the groups with different peak estradiol levels. Histogram of LEfSe analysis for vaginal microbiome across above groups. Linear discriminant analysis (LDA) scores greater than 3 are presented (NC: natural cycle group; HRT: hormone replacement cycle group; LE: low-estradiol group; ME: median-estradiol group; HE: high-estradiol group)

Table 2 Comparison of vaginal microbiota in the groups with different estradiol levels

Relative abundance (%)	NC (n = 15)	HRT (n = 10)	LE (n = 16)	ME (n = 26)	HE (n = 25)	p value
g_Escherichia-Shigella	0.15 (0.09-0.28) ^{ab}	0.225 (0.145-0.3425) ^a	0.215 (0.0975-1.31) ^a	0.115 (0.03-0.2325) ^b	0.13(0.03-0.175) ^b	0.027
g_Prevotella	0.04 (0.02-0.2) ^{ac}	0.025 (0.01-0.075) ^a	0.225 (0.065-0.8875) ^b	0.04 (0.02-0.1125) ^a	0.07(0.035-0.36) ^{bc}	0.003
g_Akkermansia	0.14 (0.05-0.56) ^a	0.045 (0.0275-0.0725) ^b	0.325 (0.065-0.4975) ^a	0.42 (0.145-0.5775) ^a	0.55(0.13-0.665) ^a	0.001
s_Lactobacillus_iners	2.22 (0.76-33.8) ^a	43.91 (1.295-90.98) ^b	1.895 (0.8175-30.85) ^a	22.235 (1.205-90.42) ^b	17.59(1.34-76.015) ^b	0.031
s_Streptococcus_anginosus	0.01 (0.00-0.09) ^a	0.01 (0.00-0.045) ^a	0.08 (0.035-1.6575) ^b	0.105 (0.0275-0.5175) ^b	0.10(0.02-0.345) ^b	0.003

Variables are presented as median (P25-P75) (Kruskal-Wallis test). Statistical significance is considered at $p < 0.05$. Superscript letters a, b, c are used to represent the significance of differences between groups. Different letters indicate a significant difference between the groups, while the same letter indicates no significant difference.

Table 3 Correlation of vaginal microbiota with the peak of estradiol

Relative abundance	The peak of estradiol level	
	<i>r</i>	<i>p</i> value
<i>g_Escherichia-Shigella</i>	−0.289	0.005
<i>g_Akkermansia</i>	0.346	0.001
<i>s_Streptococcus_anginosus</i>	0.232	0.026

Spearman correlation analysis was used. Statistical significance is considered at $p < 0.05$.

to 0.867 ($p = 0.002$). The AUC for *Atopobium* in predicting abortion was 0.899, with a 95% CI ranging from 0.825

to 0.973 ($p = 0.019$). The AUC for *Bifidobacterium* in predicting biochemical pregnancy was 0.876, with a 95% CI ranging from 0.789 to 0.962 ($p = 0.005$).

Discussion

The human vaginal microbiota plays a critical role in maintaining reproductive health. Therefore, we employed 16S rRNA sequencing to investigate the effects of the supraphysiological estrogenic state following IVF treatment on the female microbiome and its correlation with pregnancy outcomes. Our findings indicated that in patients undergoing fresh embryo transfer, the

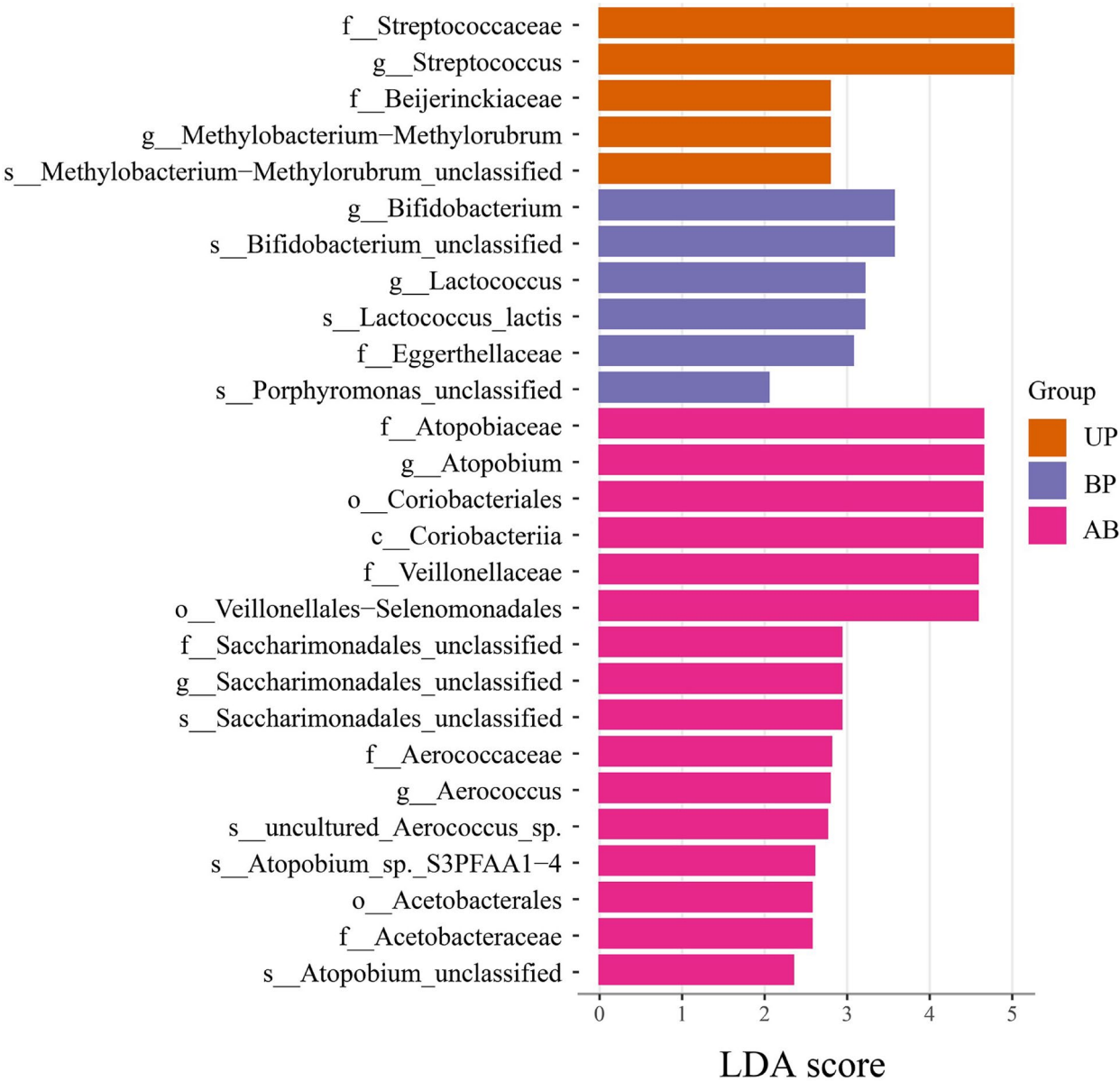


Fig. 4 Differentially abundant species across the groups with different pregnancy outcomes. Histogram of LEfSe analysis for vaginal microbiome across above groups. Linear discriminant analysis (LDA) scores greater than 2 are presented (UP: un-pregnancy group; BP: biochemical pregnancy group; AB: abortion group)

Table 4 Comparison of vaginal microbiota in the groups of different pregnancy outcomes

Relative abundance (%)	OP (n=70)	BP (n=5)	UP (n=14)	AB (n=3)	p value
<i>g_Streptococcus</i>	0.265 (0.075-0.5175) ^a	0.62 (0.32-47.07) ^{ab}	0.915 (0.39-2.63) ^b	0.46 (0.19-2.96) ^{ab}	0.003
<i>g_Atopobium</i>	0.04 (0.02-0.155) ^a	0.17 (0.015-2.145) ^{ab}	0.16 (0.02-1.67) ^{ab}	1.9 (0.4-3.02) ^b	0.040
<i>g_Bifidobacterium</i>	0.09 (0.06-0.1525) ^a	0.2 (0.16-0.3050) ^b	0.12 (0.0675-0.14) ^a	0.13 (0.11-0.22) ^{ab}	0.019

Variables are presented as median (P25-P75) (Kruskal-Wallis test). Statistical significance is considered at $p < 0.05$. Superscript letters a, b are used to represent the significance of differences between groups. Different letters indicate a significant difference between the groups, while the same letter indicates no significant difference.

dominance of *Lactobacillus* became more pronounced while microbial species diversity significantly declined with increasing estradiol levels. In contrast, in patients undergoing frozen embryo transfer, who were in a physiological estrogenic state, the dominance of *Lactobacillus* was counterbalanced by the presence of potentially pathogenic anaerobic microorganisms.

In the present study, the supraphysiological estrogenic state was found to disrupt the distribution of microbial communities. Ravel's study has demonstrated that the microbial community distribution in reproductive-age women varies significantly across different ethnic groups. In Asian women, *Lactobacillus*-dominant CSTs I, II, III, and V account for up to 80.2%, while CST IV, characterized by anaerobic bacteria dominance, accounts for 19.8% [18]. In our study, which exclusively included Chinese infertile women, we observed that among patients receiving fresh embryo transfer, those with lower peak estradiol levels (LE group) had the highest proportion of CST IV. As estradiol levels increased, the dominance of CST IV was gradually replaced by CST III. In the HE group with the highest peak estradiol levels, the microbial community distribution resembled that previously described in Asian women by Ravel et al. [18]. Among patients receiving frozen embryo transfer, the NC group with physiological estrogenic state, exhibited a CST IV dominated microbial distribution. In contrast, the CST distribution in the HRT group, which received exogenous estrogen, resembled those of the high estradiol (ME and HE) groups, with CST III being the most prevalent. These findings suggest that the CST distributions were different in infertile women under physiological hormone conditions. However, the elevated estradiol state induced by ovarian stimulation or exogenous estrogen supplementation during IVF treatment can shift the CST distributions toward those observed in most Asian women.

We also observed that there were differences in vaginal microbiota distribution, encompassing beta diversity and CST, between the NC and the HRT group with similar estradiol peak levels. The differences might be attributed to the influence of exogenous estrogen supplementation.

Previous studies generally suggest that a *Lactobacillus*-dominant microbial community serves as an indicator of optimal reproductive health in the female genital tract. A decrease in *Lactobacillus* abundance leads to a reduction in lactate production, while the metabolic byproducts of anaerobic bacteria elevate vaginal pH, thereby facilitating the colonization of opportunistic pathogens [19]. However, we found that the *Lactobacillus* dominant CST did not present a better pregnancy outcome in the supraphysiological estrogen environment. Instead, the NC group which had a higher proportion of CST IV and lacked the dominance of *Lactobacillus* exhibited better clinical pregnancy and ongoing pregnancy rates. This finding supports the hypothesis proposed by Ravel et al., which suggests that the interethnic diversity in vaginal microbiota exists, and there may not be a single "healthy" microbiota state [18]. The vaginal microbiome of infertile women might therefore differ from that of healthy women and a *Lactobacilli*-dominated vaginal microbiota did not improve pregnancy outcomes under supraphysiological estradiol conditions.

Through LEfSe analysis, we identified *Escherichia-Shigella* and *Prevotella* as biomarkers for the two groups with lower estradiol levels (HRT and LE groups). Notably, the abundance of *Escherichia-Shigella* was significantly negatively correlated with peak serum estradiol levels. The abundance of *Escherichia-Shigella* has been reported to be relatively higher in the vaginal microbiota of postmenopausal women, also demonstrating its negative correlation with the estradiol levels [20]. An elevated relative abundance of *Escherichia-Shigella* and *Prevotella* in the vaginal microbiota has been associated with a local pro-inflammatory state that affects placentation, potentially leading to repeated implantation failure, recurrent miscarriage, or preterm delivery [21]. In this study, although no significant differences in clinical outcomes were observed across the five groups, the HRT and LE groups with elevated *Escherichia-Shigella* and *Prevotella* abundance demonstrated higher non-pregnancy and lower ongoing pregnancy rates.

Akkermansia was identified as a biomarker for the HE group and exhibited a significant positive correlation with peak estradiol levels. *Akkermansia* is a Gram-negative anaerobic mucus-layer-degrading bacterium that commonly colonizes the human intestinal mucosa. It has been reported to function as an immunomodulatory probiotic in the context of autoimmune and chronic inflammatory diseases [22]. While *Akkermansia* is well-documented as a gut probiotic, it is rarely described within the context

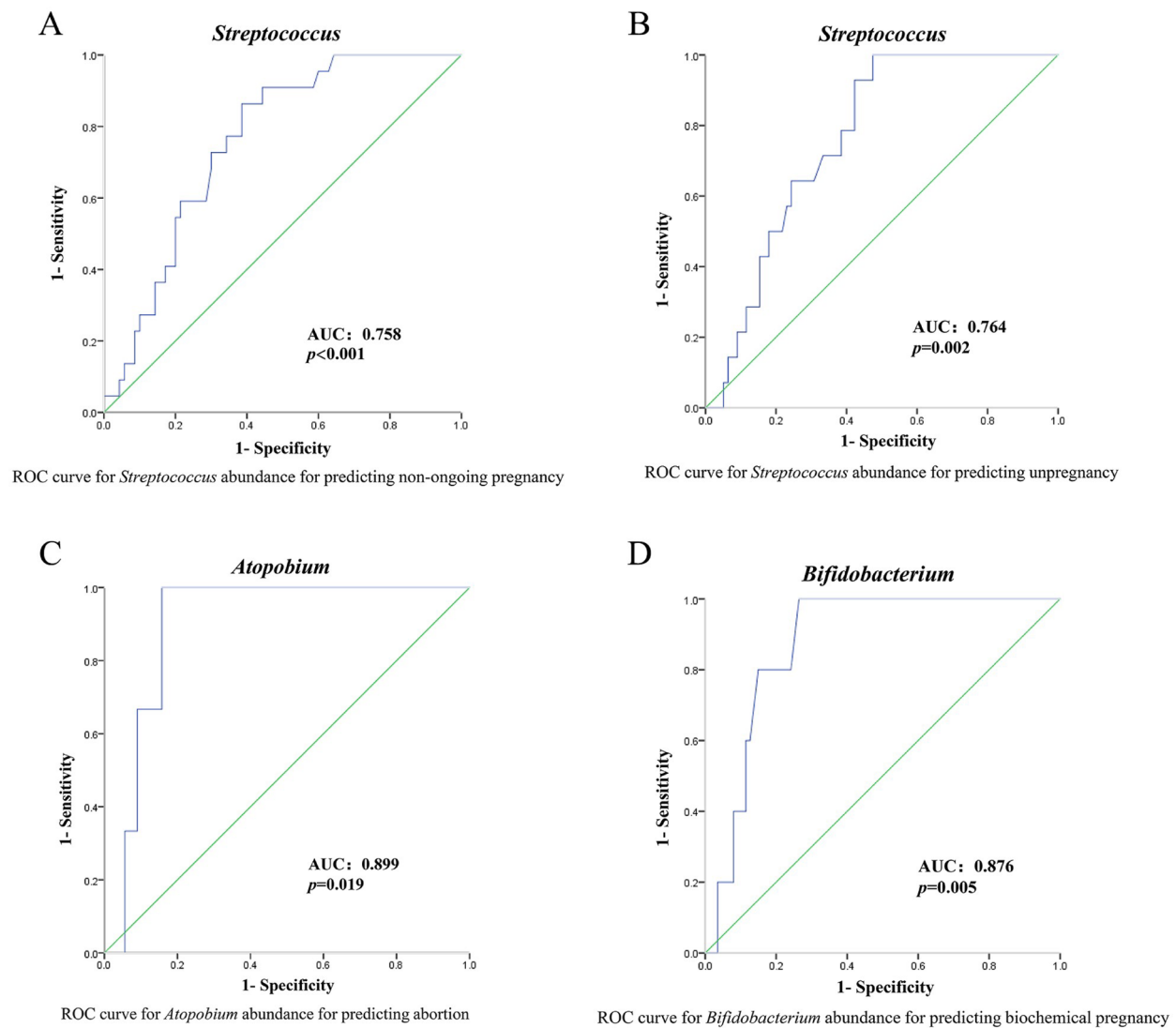


Fig. 5 ROC curves for the abundance of *Streptococcus*, *Atopobium* and *Bifidobacterium* in predicting pregnancy outcomes. **(A)** ROC curve for the abundance of *Streptococcus* in predicting non-ongoing pregnancy. **(B)** ROC curve for the abundance of *Streptococcus* in predicting non-pregnancy. **(C)** ROC curve for the abundance of *Atopobium* in predicting early abortion. **(D)** ROC curve for the abundance of *Bifidobacterium* abundance in predicting biochemical pregnancy

of the vaginal microbiota. Recent evidence revealed that vaginal colonization by *Akkermansia* may promote persistent Group B *Streptococcus* colonization, potentially extending exposure time and elevating risks of invasive perinatal and neonatal infections [23]. No significant correlation was found between *Akkermansia* abundance and IVF outcomes in the present study. However, to our knowledge, it is the first to report a correlation between the abundance of vaginal *Akkermansia* and serum estradiol levels. Furthermore, we observed that as estradiol levels increased, the abundance of vaginal *Akkermansia* changed in a similar manner with *Streptococcus anginosus*, suggesting an interaction between the two bacteria, as described before [23]. These findings provide a novel perspective on the potential role of *Akkermansia* within

the vaginal microbiome. Further investigation is needed to elucidate its function and underlying mechanisms in this context.

Lactobacillus iners was identified as a biomarker for the HRT group, with its abundance significantly higher compared to other groups. *Lactobacillus iners* is one of the common bacteria colonizing in the female reproductive tract, dominating in CSTIII [18]. Its abundance was reported to be significantly higher in women who conceived through IVF compared to those with spontaneous conception [24]. *Lactobacillus iners* is a strain of *Lactobacillus* that has a limited ability to neutralize pathogenic infections because of its shortage of producing D-lactic acid and hydrogen peroxide, in addition to low levels of antimicrobial peptides and diminished capacity

for epithelial cell adhesion [2, 3, 24]. Therefore, the lower pregnancy rate observed in the HRT group is likely attributed to the predominance of *Lactobacillus_iners* in its vaginal microbiome composition.

The abundance of *Streptococcus_anginosus* was found to be positively correlated with the peak of estradiol. In the fresh embryo transfer groups (LE, ME, HE groups) with the supraphysiologically elevated estradiol levels, its abundance was significantly higher than that in the frozen embryo transfer groups (NC, HRT groups) where the estradiol level was close to the physiological state. *Streptococcus* has been reported to be associated with poorer pregnancy outcomes [8]. We also found that *Streptococcus* was a good predictor for women who did not achieve pregnancy or ongoing pregnancy. Its abundance was significantly elevated in the UP group. The ongoing pregnancy rate was higher in the NC group with physiological estradiol level and lower abundance of *Streptococcus_anginosus*. Thus, the *Streptococcus* is highly possible a microbial mediator which interferes with pregnancy outcomes in the state of supraphysiological estrogen, but large-scale studies are needed for further confirmation.

Besides *Streptococcus*, our study still identified certain microorganisms associated with poorer pregnancy outcomes. *Atopobium* was linked to early abortion, and *Bifidobacterium* was associated with biochemical pregnancies. Infertile women undergoing IVF have been reported to exhibit significant elevation of vaginal *Atopobium* and *Bifidobacterium*, concomitant with a reduction in *Lactobacillus* [25]. Vaginal dysbiosis with increased microbial diversity, particularly the presence of *Atopobium* and *Escherichia coli* has been implicated as a contributor to adverse pregnancy outcomes [8, 14, 26, 27]. These findings suggest that the vaginal microbiome-associated biomarkers at the time of embryo transfer may offer a promising alternative approach for predicting different pregnancy outcomes.

There are several limitations to the current study. First, the cross-sectional nature of the study design resulted in an imbalance in the general clinical characteristics among the participant groups. The differences in the proportion of AMH levels and DOR in the infertile etiology were primarily attributed to the classification of patients based on peak estradiol levels, which are directly related to ovarian reservation. The disparity in the duration of infertility can be explained by the fact that most patients in the fresh embryo transfer group were undergoing their first IVF/ICSI cycle, whereas the frozen embryo transfer group included patients who had previously delivered and were undergoing a second embryo transfer. Second, we included only patients who received fresh embryo transfer following controlled ovarian stimulation with antagonist regimen, as well as those who underwent frozen embryo transfer after natural or hormone replacement

cycle endometrial preparation. Consequently, the impact of GnRH agonists induced pituitary downregulation on the vaginal microbiome could not be further explored. Third, the relatively small sample size may limit the statistical power and generalizability of the findings.

Conclusion

In Chinese infertile women undergoing IVF treatment, the distribution of vaginal microbiome on the day of embryo transfer in a physiological estradiol state differs from that previously observed in Asian women. The elevated estradiol levels induced by ovarian stimulation or exogenous estrogen supplementation could alter vaginal microbiota and shift the vaginal CSTs toward more pronounced *Lactobacillus*-dominant patterns typically seen in most Asian women, but do not help improve assisted reproductive outcomes. The abundance of *Streptococcus*, *Atopobium*, and *Bifidobacterium* on the day of embryo transfer may serve as potential biomarkers for predicting adverse pregnancy outcomes. Given the positive correlation between the abundance of *Streptococcus_anginosus* and the peak of estradiol, *Streptococcus* may act as a microbial mediator that interferes with pregnancy outcomes under supraphysiological estrogen conditions (Fig. 6). While our findings suggest potential trends, they may not fully represent the broader population. Thus, larger, prospective studies are required to provide a more thorough evaluation and to elucidate the underlying mechanisms.

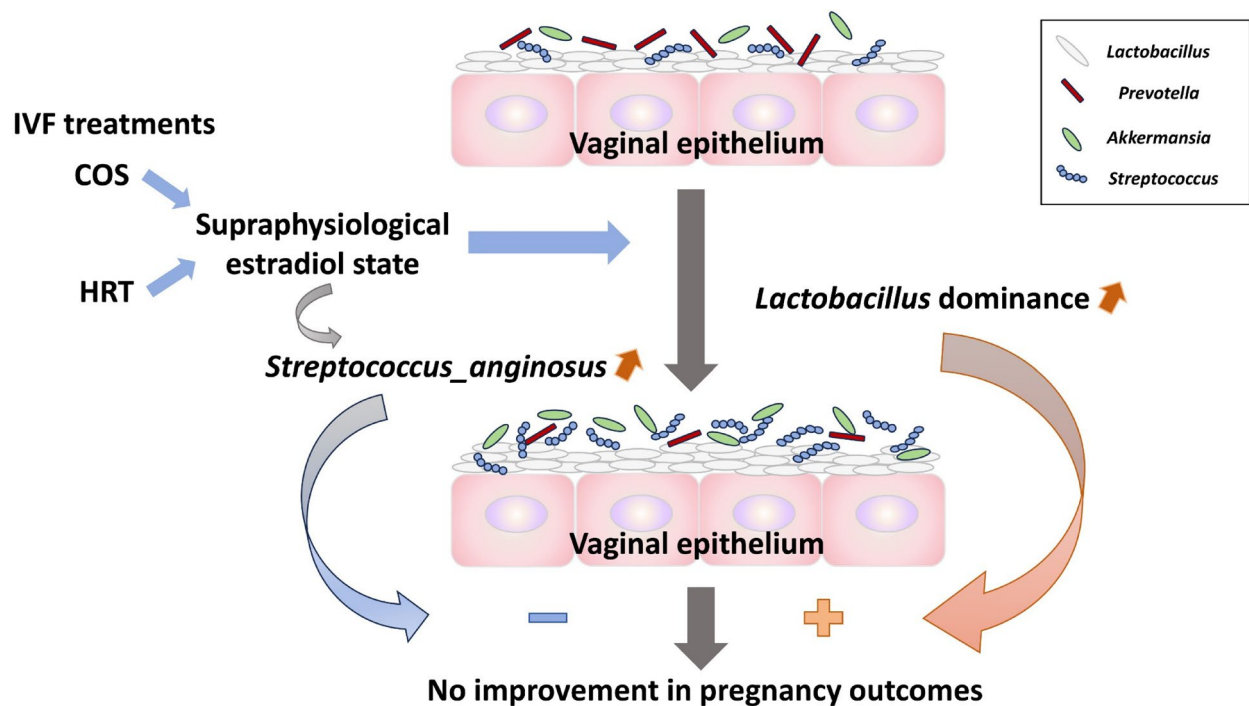


Fig. 6 Schematic view of our important findings. Although supraphysiological estradiol levels—whether induced by controlled ovarian stimulation or exogenous estrogen supplementation during IVF—may drive a more pronounced shift toward *Lactobacillus*-dominant vaginal microbiota, this change does not translate into improved pregnancy outcomes. Notably, the abundance of *Streptococcus* positively correlates with peak estradiol levels and has been linked to adverse pregnancy outcomes, suggesting that *Streptococcus* may act as a microbial mediator impairing reproductive success under hyper-estrogenic conditions. (IVF: in vitro fertilization; COS: controlled ovarian stimulation; HRT: hormone replacement)

Abbreviations

IVF-ET	In vitro fertilization-embryo transfer
HE	High-estradiol
ME	Median-estradiol
LE	Low-estradiol
NC	Natural cycle
HRT	Hormone replacement cycle
CST	Community state type
AUC	Area under the curve
E ₂	Estradiol
COS	Controlled ovarian stimulation
FET	Frozen-thawed embryo transfer
ART	Assisted reproductive technology
ICSI	Intracytoplasmic sperm injection
GnRH	Gonadotropin-releasing hormone
hCG	Human chorionic gonadotropin
PCoA	Principal coordinates analysis
LDA	Linear discriminant analysis
LEfSe	Linear discriminant analysis effect size
ROC	Receiver operating characteristic curves
BMI	Body mass index
DOR	Diminished ovarian reserve
AMH	Anti-müllerian hormone
OP	Ongoing pregnancy
BP	Biochemical pregnancy
UP	Un-pregnancy
AB	Abortion
CI	Confidence interval

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12866-025-04242-7>.

Supplementary Material 1

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Not applicable.

Author contributions

XJC contributed to the conception and design of the study, sample collection, data analysis and manuscript writing. FC contributed to data collection, data analysis and manuscript preparation. SFW contributed to sample collection. PPL contributed to statistical analysis. PL contributed to the conception and design of the study, and the revision of the manuscript. All authors approved the final manuscript and agreed to be accountable for all aspects of the work.

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Data availability

The datasets generated and analyzed during the current study are available via NCBI under the project number PRJNA1268381 (<https://www.ncbi.nlm.nih.gov/sra/PRJNA1268381>).

Declarations

Ethics approval and consent to participate

The study has been carried out in compliance with the Helsinki Declaration and was approved by the ethics review board of the Women's Hospital, School of Medicine, Zhejiang University. Informed consent was obtained from participants before sample and data collection.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Clinical trial number

Not applicable.

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References

- Das S, Bhattacharjee MJ, Mukherjee AK, Khan MR. Recent advances in understanding of multifaceted changes in the vaginal microenvironment: implications in vaginal health and therapeutics. *Crit Rev Microbiol*. 2023;49(2):256–82. <https://doi.org/10.1080/1040841X.2022.2049696>
- Schoenmakers S, Laven J. The vaginal microbiome as a tool to predict IVF success. *Curr Opin Obstet Gynecol*. 2020;32(3):169–78. <https://doi.org/10.1097/GCO.0000000000000626>
- Amabebe E, Anumba DOC. The vaginal microenvironment: the physiologic role of lactobacilli. *Front Med (Lausanne)*. 2018;5:181. <https://doi.org/10.3389/fmed.2018.00181>
- Hu XL, Feng C, Lin XH, Zhong ZX, Zhu YM, Lv PP, et al. High maternal serum estradiol environment in the first trimester is associated with the increased risk of small-for-gestational-age birth. *J Clin Endocrinol Metab*. 2014;99(6):2217–24. <https://doi.org/10.1210/jc.2013-3362>
- Kalra SK, Ratcliffe SJ, Coutifaris C, Molinaro T, Barnhart KT. Ovarian stimulation and low birth weight in newborns conceived through in vitro fertilization. *Obstet Gynecol*. 2011;118(4):863–71. <https://doi.org/10.1097/AOG.0b013e31822be65f>
- Krog MC, Hugerth LW, Fransson E, Bashir Z, Nyboe Andersen A, Edfeldt G, et al. The healthy female microbiome across body sites: effect of hormonal contraceptives and the menstrual cycle. *Hum Reprod*. 2022;37(7):1525–43. <https://doi.org/10.1093/humrep/deac094>
- Haahr T, Humaidan P, Elbaek HO, Alsberg B, Laursen RJ, Rygaard K, et al. Vaginal microbiota and in vitro fertilization outcomes: development of a simple diagnostic tool to predict patients at risk of a poor reproductive outcome. *J Infect Dis*. 2019;219(11):1809–17. <https://doi.org/10.1093/infdis/jiy744>
- Koedooder R, Singer M, Schoenmakers S, Savelkoul PHM, Morre SA, de Jonge JD, et al. The vaginal microbiome as a predictor for outcome of in vitro fertilization with or without intracytoplasmic sperm injection: a prospective study. *Hum Reprod*. 2019;34(6):1042–54. <https://doi.org/10.1093/humrep/dez065>
- Hyman RW, Herndon CN, Jiang H, Palm C, Fukushima M, Bernstein D, et al. The dynamics of the vaginal microbiome during infertility therapy with in vitro fertilization-embryo transfer. *J Assist Reprod Genet*. 2012;29(2):105–15. <https://doi.org/10.1007/s10815-011-9694-6>
- Sunkara SK, Rittenberg V, Raine-Fenning N, Bhattacharya S, Zamora J, Coomarasamy A. Association between the number of eggs and live birth in IVF treatment: an analysis of 400 135 treatment cycles. *Hum Reprod*. 2011;26(7):1768–74. <https://doi.org/10.1093/humrep/der106>
- Steward RG, Lan L, Shah AA, Yeh JS, Price TM, Goldfarb JM, et al. Oocyte number as a predictor for ovarian hyperstimulation syndrome and live birth: an analysis of 256,381 in vitro fertilization cycles. *Fertil Steril*. 2014;101(4):967–73. <https://doi.org/10.1016/j.fertnstert.2013.12.026>
- Joo BS, Park SH, An BM, Kim KS, Moon SE, Moon HS. Serum estradiol levels during controlled ovarian hyperstimulation influence the pregnancy outcome of in vitro fertilization in a concentration-dependent manner. *Fertil Steril*. 2010;93(2):442–6. <https://doi.org/10.1016/j.fertnstert.2009.02.066>
- Wei D, Zhang B, Shi Y, Zhang L, Zhao S, Du Y, et al. Effect of preconception impaired glucose tolerance on pregnancy outcomes in women with polycystic ovary syndrome. *J Clin Endocrinol Metab*. 2017;102(10):3822–9. <https://doi.org/10.1210/jc.2017-01294>
- Fu M, Zhang X, Liang Y, Lin S, Qian W, Fan S. Alterations in vaginal microbiota and associated metabolome in women with recurrent implantation failure. *mBio*. 2020;11(3):e03242–19. <https://doi.org/10.1128/mBio.03242-19>
- Zhao Y, Liao Y, Xu G, Wang Y. Endometrial microbiota alteration in female patients with endometrial polyps based on 16S rRNA gene sequencing analysis. *Front Cell Infect Microbiol*. 2024;14:1351329. <https://doi.org/10.3389/fcimb.2024.1351329>
- Bolyen E, Rideout JR, Dillon MR, Bokulich NA, Abnet CC, Al-Ghalith GA, et al. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. *Nat Biotechnol*. 2019;37(8):852–7. <https://doi.org/10.1038/s41587-019-0209-9>
- Segata N, Izard J, Waldron L, Gevers D, Miropolsky L, Garrett WS, et al. Metagenomic biomarker discovery and explanation. *Genome Biol*. 2011;12(6):R60. <https://doi.org/10.1186/gb-2011-12-6-r60>
- Ravel J, Gajer P, Abdo Z, Schneider GM, Koenig SS, McCulle SL, et al. Vaginal microbiome of reproductive-age women. *Proc Natl Acad Sci U S A*. 2011;108(1 Suppl 1):4680–7. <https://doi.org/10.1073/pnas.1002611107>
- Biagi E, Vitali B, Pugliese C, Candela M, Donders GG, Brigid P. Quantitative variations in the vaginal bacterial population associated with asymptomatic infections: a real-time polymerase chain reaction study. *Eur J Clin Microbiol Infect Dis*. 2009;28(3):281–5. <https://doi.org/10.1007/s10096-008-0617-0>
- Lan Y, Jin B, Zhang Y, Huang Y, Luo Z, Su C, et al. Vaginal microbiota, menopause, and the use of menopausal hormone therapy: a cross-sectional, pilot study in Chinese women. *Menopause*. 2024;31(11):1014–23. <https://doi.org/10.1097/GME.0000000000002432>
- Carosso A, Revelli A, Gennarelli G, Canosa S, Cosma S, Borella F, et al. Controlled ovarian stimulation and progesterone supplementation affect vaginal and endometrial microbiota in IVF cycles: a pilot study. *J Assist Reprod Genet*. 2020;37(9):2315–26. <https://doi.org/10.1007/s10815-020-01878-4>
- Rodrigues VF, Elias-Oliveira J, Pereira IS, Pereira JA, Barbosa SC, Machado MSG, et al. Akkermansia muciniphila and gut immune system: A good friendship that attenuates inflammatory bowel disease, obesity, and diabetes. *Front Immunol*. 2022;13:934695. <https://doi.org/10.3389/fimmu.2022.934695>
- Burcham LR, Burcham ZM, Akbari MS, Metcalf JL, Doran KS. Interrelated effects of zinc deficiency and the microbiome on group B Streptococcal vaginal colonization. *mSphere*. 2022;7(4):e0026422. <https://doi.org/10.1128/msphere.00264-22>
- Lledo B, Fuentes A, Lozano FM, Cascales A, Morales R, Hortal M, et al. Identification of vaginal microbiome associated with IVF pregnancy. *Sci Rep*. 2022;12(1):6807. <https://doi.org/10.1038/s41598-022-10933-2>
- Zhao C, Wei Z, Yang J, Zhang J, Yu C, Yang A, et al. Characterization of the vaginal microbiome in women with infertility and its potential correlation with hormone stimulation during in vitro fertilization surgery. *mSystems*. 2020;5(4):e00450–20. <https://doi.org/10.1128/mSystems.00450-20>
- Donders GG, Van Calsteren K, Bellen G, Reybrouck R, Van den Bosch T, Riphagen I, et al. Predictive value for preterm birth of abnormal vaginal flora, bacterial vaginosis and aerobic vaginitis during the first trimester of pregnancy. *BJOG*. 2009;116(10):1315–24. <https://doi.org/10.1111/j.1471-0528.2009.02237.x>
- Hyman RW, Fukushima M, Jiang H, Fung E, Rand L, Johnson B, et al. Diversity of the vaginal microbiome correlates with preterm birth. *Reprod Sci*. 2014;21(1):32–40. <https://doi.org/10.1177/1933719113488838>

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